Data Collection and Preprocessing Phase

Date	17 July 2024
Team ID	xxxxxx
Project Title	WCE Curated Colon Disease Classification
Maximum Marks	2 Marks

Data Collection Plan & Raw Data Sources Identification Template

The model requires labelled data consisting of Colonoscopy Images and associated labels for each image indicating the presence or absence of colon disease and the specific disease type. Such data can be collected from medical institutions, public databases and research collaborators.

Data Collection Plan Template

Section	Description
Project Overview	The project aims to classify the WCE Curated colon diseases based on Wireless Capsule Endoscopy (WCE) images. The project aims to reduce time in detection and classification of the disease.
Data Collection Plan	Data will be collected from an already existing dataset on Kaggle.
Raw Data Sources Identified	Data can be collected from medical institutions, public databases and research collaborators.

Raw Data Sources Template

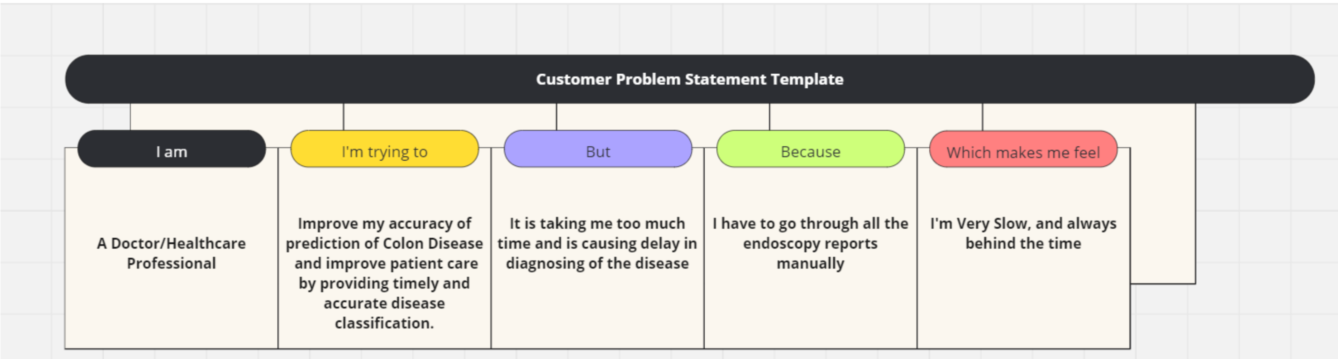
Source Name	Description	Location/URL	Format	Size	Access Permissions
WCE Curated Colon Disease Dataset Deep Learning	The dataset is taken from J. Silva, A. Histace, O. Romain, X. Dray and B. Granado, "Toward embedded detection of polyps in WCE images for early diagnosis of colorectal cancer", International Journal of Computer Assisted Radiology and Surgery, vol. 9, no. 2, pp. 283-293, 2013. DOI:10.1007/s11548-013-0926-3.	https://www.kaggle.com/datasets/francismon/curated-colon-dataset-for-deep-learning	Images	2 GB	Public

Project Initialization and Planning Phase

Date	11 July 2024
Team ID	SWTID1720011518
Project Name	WCE curated Colon Disease Classification using Deep Learning
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Create a deep learning model to classify colon diseases using medical imaging data in a curated manner. This initiative intends to reliably categorize diverse colon disorders through the analysis of patient data and colonoscopy images, therefore facilitating early identification, treatment planning, and improved patient outcomes.
Use the deep learning model to aid in the diagnosis of colon disorders by medical practitioners. By providing rapid and precise illness categorization, we may enhance patient care, expedite treatment decisions, and increase diagnostic accuracy.



Reference: https://miro.com/app/board/uXjVKzylESg=/?share_link_id=115441034671

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A Doctor/Healthcare Professional	Improve my accuracy of prediction of Colon Disease and improve patient care by providing timely and accurate disease classification.	It is taking me too much time and is causing delay in diagnosing of the disease	I have to go through all the endoscopy reports manually	I'm Very Slow, and always behind the time

Project Initialization and Planning Phase

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Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To assist healthcare professionals in diagnosing colon diseases. Enhance diagnostic accuracy, streamline treatment decisions, and improve patient care by providing timely and accurate disease classification.
Scope	The scope involves collecting and preprocessing high-quality colonoscopy images, developing a robust deep learning model, integrating it into healthcare systems, and ensuring compliance with ethical and regulatory standards.
Problem Statement	
Description	Healthcare professionals struggle with accurately diagnosing colon diseases due to inconsistent imaging data and complex medical records. Developing a deep learning model to analyze these data sources can enhance diagnostic precision, support early detection, and improve treatment planning and patient outcomes.
Impact	The solution will significantly improve diagnostic accuracy, enabling early detection of colon diseases and facilitating timely interventions. It will streamline treatment planning, providing healthcare professionals with reliable data for informed decisions. This will enhance patient outcomes and operational efficiency, reducing the workload on medical staff and improving overall patient care.

Proposed Solution	
Approach	We will utilize a Kaggle dataset and employ transfer learning for this project. By extracting features using various pre-trained models, we will evaluate their performance and select the model that yields the best results. This approach ensures optimal accuracy and efficiency in classifying colon diseases from medical imaging data.
Key Features	The key features of the proposed solution include using a comprehensive Kaggle dataset, employing transfer learning for efficient feature extraction, evaluating multiple pre-trained models for optimal performance, and integrating the best-performing model into healthcare systems. This approach enhances diagnostic accuracy, supports early detection, and streamlines treatment planning for colon diseases.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Any Basic GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	512 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Tensorflow, Numpy
Development Environment	IDE, version control	Google Colab, Spyder
Data		
Data	Source, size, format	Kaggle dataset (WCE Curated Colon Disease Dataset Deep Learning), 6,000 images